

Task 1 — HMI & Laptop Vision Requirements

Industry-grade Software System Requirements Specification (SRS)

Project: FOG-ORCHESTRATOR 2.0 | Task 1 — HMI & Optional Laptop Computer Vision

1. Purpose and Scope

Task 1 delivers the operator/control-room Human–Machine Interface (HMI) for FOG-ORCHESTRATOR 2.0. The HMI is mandatory for the SIH prototype. Computer vision is optional and, if implemented, shall execute on the teammate's laptop/central workstation—not on the ESP32 or vehicle edge controller. The HMI shall visualize live fleet state, road state, fog/visibility state, safety envelopes, bottlenecks, dispatch decisions, communication health, and alerts, while remaining clearly separated from the safety-critical vehicle controller.

2. System Boundary

In Scope	Out of Scope
Control-room / operator HMI	Real mine FMS certification
Digital-twin data visualization	Direct OEM HEMM drive-by-wire certification
Fleet status and road-state visualization	Autonomous steering
Safe speed / headway / risk display	Safety-certified alarm hardware
Dispatch / slot command visualization and acknowledgment	On-device neural inference
Optional laptop-side computer vision	Field-deployed production system
Data logging, playback and event audit	Replacement of NMDC FMS

3. Operational Concept

- HMI is the human-facing supervisory layer. It does not replace the Tier-1 local physics safety governor.
- The digital system may recommend or issue target speed, departure slot, route, or hold commands; the vehicle controller must enforce local safety limits.
- When central communication is unavailable, HMI shall indicate degraded mode; the vehicle shall continue local-safe operation.
- Optional computer vision is an information source for visibility/obstacle context and shall not directly override physics constraints.

4. User Roles

Role	Primary Need	HMI Permissions
Vehicle operator	Immediate safe speed, headway, risk, slot/route guidance	View vehicle page; acknowledge warnings
Control-room operator	Fleet-wide state, bottlenecks, dispatch, incidents	Fleet/road view; dispatch recommendations; acknowledge/override non-safety advisory actions
Safety/engineering reviewer	Traceability, event history, replay, KPIs	Read-only analytics, logs, scenario playback
Developer/tester	Telemetry debugging, health and data quality	Diagnostic panels; simulator controls

5. Functional Requirements

ID	Requirement	Statement
HMI-FR-001	Live system overview	HMI shall show current system mode, timestamp, connectivity status, active alerts, and fleet count.
HMI-FR-002	Mine/road map	HMI shall render the graph-based digital mine map with road segments, grades, intersections, switchbacks, shovels, crushers and vehicle positions.
HMI-FR-003	Vehicle state	For each vehicle, show ID, position/segment, speed, safe speed, headway, risk state, friction estimate (with confidence), and communication state.
HMI-FR-004	Safety envelope	HMI shall clearly distinguish actual speed from v_{safe} and identify the active limiting constraint (visibility, friction, grade, braking/retarder, curve or site limit).
HMI-FR-005	Headway monitoring	HMI shall show current headway and required H_{safe} for selected vehicle pairs; violation status shall be visible.
HMI-FR-006	Bottleneck visualization	HMI shall rank and visually mark bottlenecks using BottleneckScore and/or utilization, queue length and criticality.
HMI-FR-007	Queue monitoring	HMI shall display crusher/shovel/intersection queues, queue limits, arrival rate and service rate where available.
HMI-FR-008	Arrival-rate shaping	HMI shall show planned vs actual arrivals at critical nodes and any active metering/holding decision.
HMI-FR-009	Slot reservation	HMI shall show switchback/intersection reservations, ETA, slot window, current holder, and conflict status.
HMI-FR-010	Dispatch decision	HMI shall show recommended/issued vehicle assignment, route and departure slot with reason code.
HMI-FR-011	Fog/visibility	HMI shall show current visibility and, when available, forecast visibility with uncertainty/confidence band.
HMI-FR-012	Road condition	HMI shall show friction estimate, road wetness/surface state and optional roughness state where available.
HMI-FR-013	Communication health	HMI shall show V2V/V2I/LoRa link state, packet freshness/age and degraded-mode state.
HMI-FR-014	Alerts	HMI shall generate prioritized safety/operational alerts including unsafe speed, unsafe headway, bottleneck risk, communication loss, stale data and slot conflict.
HMI-FR-015	Alert acknowledgment	Operators shall be able to acknowledge applicable advisory/operational alerts; safety-critical local protection shall not depend on acknowledgment.
HMI-FR-016	System modes	HMI shall show NORMAL, CAUTION, DEGRADED, LOCAL-SAFE and STOP/UNSAFE modes.
HMI-FR-017	Replay	HMI shall support time-based replay of logged simulation/HIL telemetry for post-event analysis.
HMI-FR-018	KPI dashboard	HMI shall show throughput, cycle time, queue length, utilization, stops, safety-envelope violations and recovery time.
HMI-FR-019	Scenario controls	Developer/test view shall permit loading named fog, friction, grade, fleet-density and communication-loss scenarios.
HMI-FR-020	CV integration (optional)	If CV is implemented, HMI shall display camera frame, visibility estimate/confidence and detected obstacle/road-state overlays without placing inference on the ESP32.

6. Non-Functional Requirements

ID	Category	Requirement
HMI-NFR-001	Performance	Primary dashboard update target: ≤ 1 s for live state refresh under prototype load.
HMI-NFR-002	Command latency	Display of a new central decision target: ≤ 2 s from receipt under nominal LAN conditions; measured separately from vehicle actuation latency.
HMI-NFR-003	Data freshness	Each live datum shall carry timestamp/source age; stale data shall be visually flagged after configurable timeout.
HMI-NFR-004	Availability	HMI should recover automatically after temporary backend/reconnect interruption without corrupting state.
HMI-NFR-005	Safety separation	Loss or crash of HMI shall not disable Tier-1 vehicle safety.
HMI-NFR-006	Explainability	Every dispatch/risk decision shown to the operator shall include a concise reason code and key limiting variables.
HMI-NFR-007	Traceability	All issued/received commands and mode transitions shall be timestamped and logged.
HMI-NFR-008	Usability	Critical information shall be readable at a glance; avoid relying on color alone for safety state.
HMI-NFR-009	Scalability	Prototype HMI shall support at least 50 simulated vehicles and 20 physical/logical road nodes.
HMI-NFR-010	Portability	Laptop deployment shall work from documented setup steps and use open-

ID	Category	Requirement
		source tooling where practical.
HMI-NFR-011	Security baseline	Prototype communications shall include authentication/configurable access controls where supported; no hard-coded credentials in source.
HMI-NFR-012	Recovery	If backend data stops, the HMI shall enter DATA-Stale/DEGRADED indication rather than freezing values silently.

7. Minimum HMI Screens

Screen	Mandatory Content
S1 — Operations Overview	Mine map, vehicles, active bottleneck, current fog state, top alerts, throughput and system mode.
S2 — Vehicle Detail	Speed, safe speed, headway, risk, friction, grade, communication, active constraint, recent trend.
S3 — Bottleneck & Queue	Crusher/shovel/intersection utilization, arrival/service rates, queue lengths, predicted queue trend.
S4 — Dispatch & Slots	Truck assignment, departure time, route, switchback/intersection slot timeline, reason codes.
S5 — Event/Replay	Timeline of fog, alerts, commands, queue changes, violations and recovery.
S6 — Diagnostics	Data freshness, sensor/telemetry status, backend health, message counts, latency.

8. Optional Laptop Computer Vision

Computer vision is optional for Task 1 and runs on the teammate laptop/central workstation. It shall be treated as a non-safety-critical perception aid. It may provide: visibility estimation from contrast/haze metrics; obstacle/vehicle detection; road wetness/surface classification; and visual overlays. The CV subsystem shall publish a timestamped estimate plus confidence. The physics/safety engine remains authoritative.

CV ID	Requirement	Acceptance
CV-001	Camera ingestion	Read live webcam/IP stream or recorded video.
CV-002	Visibility estimate	Produce an interpretable visibility/scene-visibility proxy plus confidence.
CV-003	Obstacle/vehicle detection	Optional detector shall identify configured classes in test footage.
CV-004	Latency	Target ≤ 200 ms from frame capture to published CV result on test laptop; report measured value.
CV-005	Graceful failure	Camera failure shall set CV status to unavailable; no stale output may be silently treated as current.
CV-006	Integration	Publish results to HMI/backend with timestamp, source ID and confidence.

9. Data Interfaces

Message	Minimum Fields
VehicleState	vehicle_id, timestamp, x/y or segment_id, speed_mps, accel_mps2, grade_rad, friction_est, friction_sigma, mode, comm_confidence
SafetyState	vehicle_id, v_safe, h_safe, actual_speed, active_constraint, risk_level
RoadState	segment_id, visibility_m, visibility_sigma, friction, grade, capacity_vph, queue, utilization
BottleneckState	node_id, lambda_vph, mu_vph, queue, queue_max, criticality, bottleneck_score
DispatchCommand	vehicle_id, route_id, departure_time, target_speed, slot_id, reason_code, timestamp, command_id
SlotState	slot_id, resource_id, vehicle_id, start_time, end_time, status
CVResult	timestamp, source_id, visibility_proxy, confidence, detections[], road_state
Health	component_id, timestamp, state, latency_ms, age_ms, error_code

10. Technology Baseline

- Prototype frontend: web-based HMI is preferred for fast demonstration and platform portability (e.g., React/HTML/JS or Streamlit if time-constrained).
- Backend/API: Python service (e.g., FastAPI/Flask) or direct integration with the digital-twin process.
- Real-time transport: WebSocket/MQTT/local message bus; final choice shall be documented.
- Charts: Plotly/Matplotlib-compatible exported data is acceptable.
- Computer vision: laptop-side Python/OpenCV and an approved detector if used; no ESP32 inference required.

11. Requirement Traceability Principle

Every HMI function shall have a unique requirement ID, an implementation reference, and at least one verification method. Requirements marked mandatory must not be waived without team-lead approval. Optional CV requirements can be marked N/A without affecting HMI completion.

12. Task 1 Completion Definition

- Mandatory HMI screens implemented and connected to live/simulated data.
- All mandatory HMI functional requirements verified.
- Stale-data and communication-loss behavior demonstrated.
- Safety-critical values (v_safe, h_safe, risk, active constraint) visible and traceable.
- Bottleneck, queue, arrival shaping and slot reservation visible.
- Logs and replay available.
- Optional CV may be separately marked Complete / Partial / Not Implemented.